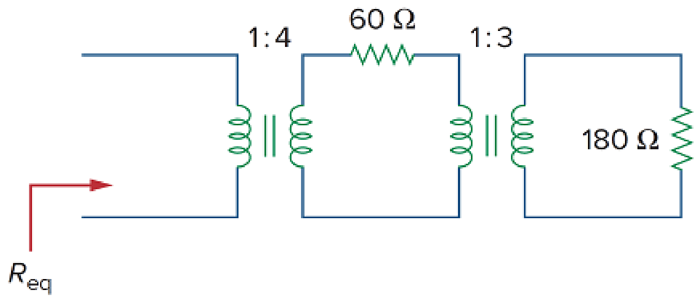
Chapter 13, Problem 55P

Problem

For the circuit in Fig. 13.120, calculate the equivalent resistance.

Figure 13.120



Step-by-step solution

Step 1 of 2

Refer to Figure 13.120 in the textbook for a transformer circuit.

Use the reflected impedance concept to find the equivalent resistance of the circuit.

Observe that there are two transformers in the circuit. First, reflect the secondary resistance of second transformer to the primary side.

The secondary resistance of the second transformer is,

$$R_3 = 180\ \Omega$$

The reflected resistance of the second transformer is,

$$R_{32} = \frac{R_3}{n_2^2}$$

Here, n_2 is the turns ratio of second transformer.

$$\begin{aligned} n_2 &= \frac{N_{22}}{N_{12}} \\ &= \frac{3}{1} \\ &= 3 \end{aligned}$$

Therefore,

$$\begin{aligned} R_{32} &= \frac{R_3}{n_2^2} \\ &= \frac{180}{3^2} \\ &= \frac{180}{9} \\ &= 20\ \Omega \end{aligned}$$

Step 2 of 2

Calculate the secondary resistance of the first transformer.

$$\begin{aligned} R_2 &= 60 + R_{32} \\ &= 60 + 20 \\ &= 80\ \Omega \end{aligned}$$

The reflected resistance of the first transformer is,

$$R_{21} = \frac{R_2}{n_1^2}$$

Here, n_1 is the turns ratio of first transformer.

$$\begin{aligned} n_1 &= \frac{N_{21}}{N_{11}} \\ &= \frac{4}{1} \\ &= 4 \end{aligned}$$

Therefore,

$$\begin{aligned} R_{21} &= \frac{R_2}{n_1^2} \\ &= \frac{80}{4^2} \\ &= \frac{80}{16} \\ &= 5\ \Omega \end{aligned}$$

Calculate the equivalent resistance of the circuit.

$$\begin{aligned} R_{eq} &= R_{21} \\ &= 5\ \Omega \end{aligned}$$

Therefore, the equivalent resistance of the circuit R_{eq} is $\boxed{5\ \Omega}$.